



Long-Term Dynamics of Chromosomal Instability in Cancer: A Transition Probability Model

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Abstract—A stochastic model of chromosomal instability has been previously developed which has included one adjustable parameter—the probability of a segregation error. Using computer simulations, we have previously analyzed this model and were able to reproduce a short-term dynamics of chromosome copy number distributions in clones of cancer cells. In a short run, segregation errors provide a continuous production of deviant cells with increasing variation of cell karyotypes, which depends upon the rate of segregation errors. In the long-term observations, many tumors and cancer cell lines have been observed to maintain a stable, although abnormal, distribution of chromosome number for hundreds of cell generations. This phenomenon of “stability within instability” presents an interesting paradox, which could be addressed mathematically. However, this would require modeling of long term growth of tumor cell clones for hundreds of generations, which has far exceeded capabilities of modern computer systems. In this study, we have analyzed asymptotic behavior of our model using a semianalytical approach. A transition probability matrix was derived analytically and implemented in a recursive algorithm for computational experiments. Using this transition probability model, the expected frequencies of chromosome copy number have been calculated under various initial and boundary conditions. We have also tested several alternative models, which describe various mechanisms of errors in segregation of chromosomes, and found conditions for stabilization of distribution of chromosomes copy numbers over a large number of cell generations. Stable clonal frequencies were estimated which are independent of initial conditions, i.e., chromosome copy numbers in the initiator cells. These stable distributions were, however, dependent on the model assumptions regarding particular mechanism of errors in segregation of chromosomes. Thus, our modeling results have suggested a possible connection between the form of stable distribution of chromosome numbers in tumors and the underlying mechanism of errors in segregation of chromosomes. This new analytical approach allows us to overcome technical impairments and limitations of computer simulation. and, for the first time, provides mathematical insight into long-term evolution of chromosome numerical changes in human tumors. © 2001 Elsevier Science Ltd. All rights reserved.

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1. INTRODUCTION

The karyotypes (i.e., sets of chromosomes) of normal human cells are stable over a lifetime of the organism with small variation which is often increased with aging [1,2]. Although it has been shown that molecular mechanisms of chromosome-spindle attachment are prone to random error [2,3], the cells with aberrant or incomplete karyotypes normally are not viable and are quickly eliminated from the organism. However, such chromosomal aberrations, which occur often in premalignant lesions and during neoplastic transformation, may persist and propagate throughout many stages of tumor development [4–13]. This distinctive property of cancer cells has recently become a focus of many experimental studies and led to the recognition of the chromosomal instability as an important mechanism of carcinogenesis and an integral feature of human cancers [14–17]. With improvements in technique, the gross numerical and structural chromosomal aberrations were found in a majority of solid human tumors [10,18,19]. For colorectal cancers, an estimated 80% of tumors have the chromosomal instability phenotype (CIN phenotype) [20].

However, the mechanisms generating chromosome numerical changes and the long-term outcomes of such instability are not well understood. Although many genetic defects have been identified which may lead to mitotic infidelity [5,18,21–24], their role in destabilization of cell karyotype remains unclear. Importantly, most of the abnormal mitotic events lead to a failure of chromosome segregation at cell division. In the short run, such chromosome segregation errors (SE) produce numerical chromosomal abnormalities and in the long run might contribute to a long-term genetic evolution of tumors [24].

Paradoxically, many chromosomally unstable tumors and cancer cell lines have been observed to maintain a stable, although abnormal, distribution of chromosome number for hundreds of cell generations [9]. This phenomenon of “stability within instability” presents an interesting paradox, which could be addressed mathematically. Also, many important questions remain unanswered regarding the actual rates and specific mechanisms of segregation errors. In this study, we use mathematical modeling to address these questions of the long-term chromosomal changes in cancer, and a possible role of segregation errors in evolution of tumor cells karyotypes.

2. CONCEPTUAL MODEL OF CHROMOSOME SEGREGATION ERRORS

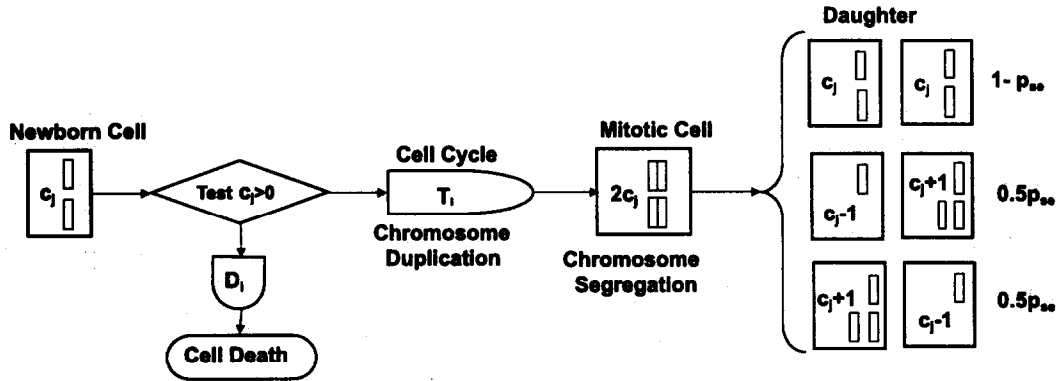
We have previously developed a mathematical model which describes a population of dividing cells each containing set of chromosomes. In the model, chromosomes are duplicated within the cells and redistributed into daughter cells at each cell division. The model describes chromosome duplication and segregation as an embedded branching process of “division-within-division” with continuous time. Detailed analysis of this model is published elsewhere [25].

Briefly, cells were described as discrete objects dividing by a binary fission. Each cell in a population contained an integer number of discrete objects—chromosomes of 23 types which undergo replication and segregation at every cell division cycle. The main assumption of the model was that chromosomes are prone to segregation errors which occur at random and affect each chromosome independently. Random numerical changes of the amount of chromosomes in the cells were described as a random walk of cells from one karyotype to another with variable “step” determined at division with given probability.

Similar models have been applied by others [26–28] to the analysis of biological processes of replication and segregation of molecular objects within the dividing cells. The examples include the processes of gene amplification [27], and telomere shortening [28]. However, previous models were concerned with specific mechanisms of single gene amplifications or telomere shortening and the outcomes described by one-dimensional branching random walks. In our model, we have described mechanisms of duplication and segregation which are specific for chromosomes and have considered simultaneous evolution of many types of chromosomes as a multi-dimensional random branching walk.

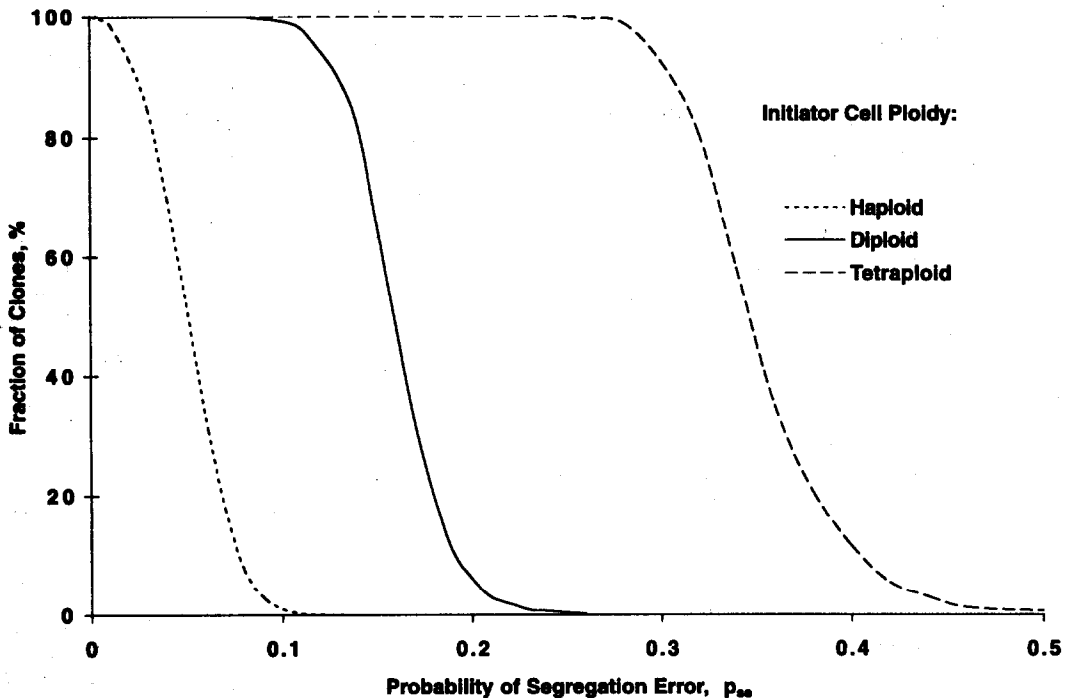
The biological assumptions of the basic model were minimal and simple. Model parameters include the number of chromosome types in a cell and distribution of time to division. For every chromosome, we describe two types of nonrandom events: duplication and segregation (Figure 1a). Model describes segregation error as a random event of transfer of both copies of chromosome into the same daughter cell. In the model, we assume that all chromosomes act independently and have equal probability of segregation error p . This single parameter reflects an average rate of segregation errors per cell division.

(a) Conceptual model.



(a) Flow diagram: cycle of cell division, chromosome duplication, and segregation. One type of chromosome shown as an example with normal diploid copy number (two copies). In the model, we assumed constant probability of chromosome segregation errors, p_{se} .

(b) Surviving fraction of clones (% per 2,500 clones).



(b) Survival fraction of single cell derived clones. Results of computer simulation show strong dependence of clone survival on the probability of segregation errors and initial conditions (chromosome number in initiator cells).

Figure 1. Conceptual model of chromosome segregation errors.

Importantly, one event of single segregation error affects both daughter cells. Each segregation error produces gain of chromosome in one daughter cell and loss of chromosomes in another daughter cell at the same division. In our model, we defined boundary conditions which limit loss and gain of chromosomes of each particular type. Our basic model reflects the biological fact that cells require some minimal set of chromosomes in order to survive. For simplicity, the minimal number of chromosomes was set to one copy for each type of chromosomes. In other words, we assumed that a cell cannot survive if it loses all chromosomes of any type (Figure 1a). These conditions of cell loss defined lower absorption boundaries of the random walk model.

Although there is no consensus on the actual number of chromosomes which are essential for cell survival, for human cells at least one chromosome copy must be present for the most of the chromosome types. In the model, we tested a range of essential chromosome numbers and obtained quantitatively similar results.

Our basic model did not include limitations on maximum number of chromosome copies which a cell could gain as a result of segregation errors. Many biological observations suggest that such limitations do exist in the range from eight to 32 copies and vary significantly for different tumors and cell lines. The impact of such upper boundary conditions is relatively small for the short-term dynamics of clonal growth but become essential for consideration of long term growth of cell clones. These additional boundary conditions were included in our current transition probability model and tested in computational experiments (see Section 4.5.1).

3. COMPUTER SIMULATION EXPERIMENTS

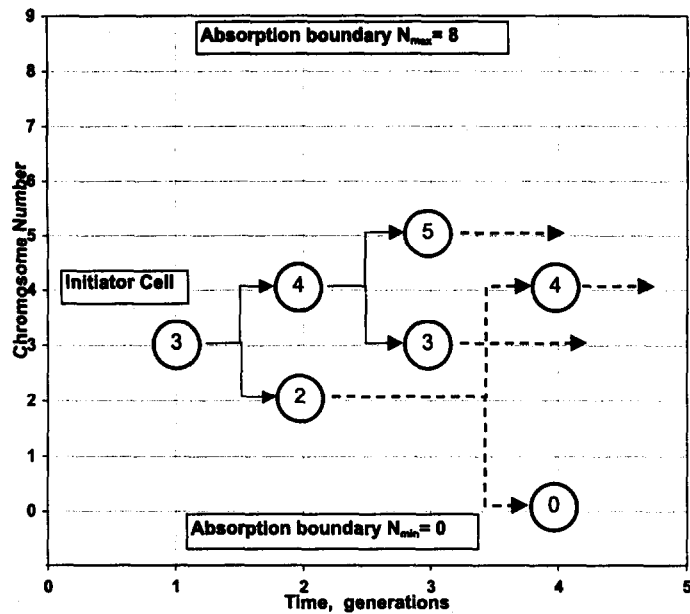
Our conceptual model was implemented in a computer simulation program. A C++ program (Microsoft Visual C++) describes a population of dividing cells using a discreet event simulation algorithm. The statistics of chromosome number distributions were estimated based on a large number of computer simulations of clones each originating from a single cell, and expanded for many cell generations. The properties of the proposed model were previously studied in a series of computer simulation experiments [25]. In all experiments, growth of clones from a single cell was simulated multiple times for each set of parameter values, initial and boundary conditions (Figure 2a). Standard pseudorandom number generators were used to generate random variables. A typical outcome of a single simulation run consisted of information about the number of cells in a clone and the number of chromosomes of each type in every cell at predefined points in time. Outcomes of multiple simulation runs for 25 generations were analyzed using descriptive statistics and were compared with experimental data on average frequencies of several chromosome types (Figure 2b).

We have estimated survival fraction of clones as a function of the probability of segregation errors. Based on survival information, we analyzed the dynamics of growth and the chromosome number distributions within the clones.

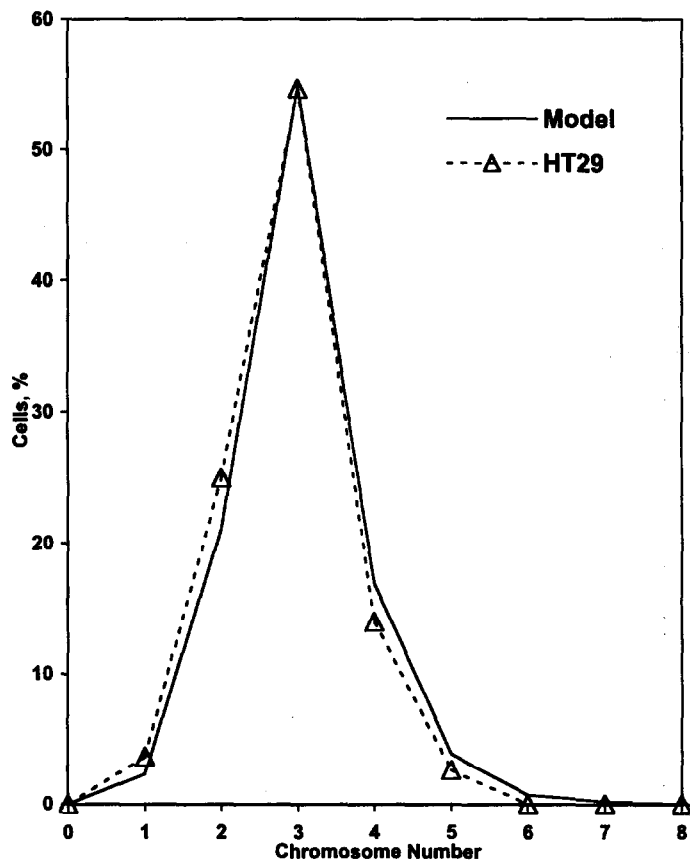
3.1. Previous Results

In summary, our previous findings suggested [25] that in the short term, the outcomes of clonal growth directly depend upon the rate of segregation errors and the initial conditions, i.e., number of chromosome copies in the initiator cell (Figure 1b).

For relatively small rates of segregation errors (SE), growth of clones is stable, 100% of clones survive, and average ploidy (total amount of chromosomes per cell) of cell population remains stable and equal to the number of chromosomes in the initiator cell. However, variance of total amount of chromosomes increases with time. As SE rate increases, clonal growth becomes unstable. The survival fraction of clones decreases inversely proportionally to SE rate. The average cell ploidy shifts toward larger total chromosome numbers per cell (ploidy increase). For the rates of SE exceeding some critical value, all clones become extinct in a few generations (Figure 1b).



(a) Growth of clones from a single initiator cell shown with the examples of possible outcomes of segregation errors. The number of chromosomes of one type is presented, although in the model we have considered all 23 types of chromosomes simultaneously.



(b) Chromosome number distribution in clones after 25 generations. Average results of simulation ($p_{se} = 0.01$) versus experimental data for cancer cell line HT29 [20].

Figure 2. Simulation experiments.

The theoretical limits were estimated for survival of clones which defined relatively narrow interval of segregation error rates (Figure 1b). These limits depend on initial conditions, i.e., the chromosomal make-up of the initiator cells. For instance, clones which were initiated from tetraploid cells (four copies of each chromosome type) can survive rate of segregation errors which is two times higher than the critical rate for the diploid initiator cells (two copies of each chromosome types, Figure 1b). Interestingly, the model suggests that the clones initiated by haploid cells (one copy of each chromosome type) have practically no plateau: even for small SE rates, the survival fraction of such clones is less than 100% (Figure 1b). With the increase of ploidy, cells could tolerate much higher rates of SE, with the "safety plateau" of 100% survival fraction expanding proportionally to the cell ploidy (Figure 1b).

3.2. Estimation of Segregation Errors Rate for Colorectal Cancer Cell Lines

In our recent study [29], we tested a hypothesis that random segregation errors could be a major mechanism of chromosomal instability in colorectal cancers cell lines. Recently published observations provided counts of 12 chromosome types within single cell derived clones of colorectal cancer cell lines [20]. These chromosome frequency distributions were satisfactorily reproduced in our computer simulation experiments for up to 25 cell generations (Figure 2b). The rates of segregation errors for each cell line were estimated and will be published elsewhere [29].

Briefly, statistics of expected frequencies were calculated on large series of "virtual" clones generated for each tested value of SE rate. The input parameters of our model have included only initial conditions of clonal growth: chromosomal content of initiator cells and constant probability of segregation errors. These estimated expected frequencies of chromosome copies were then compared with average frequencies of chromosomal content for each colorectal cell line using *t*-statistics and χ^2 -test [30,31]. The modeling results were in a good agreement with chromosomal profiles of clones which were grown for 25 generations. The estimated probabilities of segregation errors in unstable cell lines were in the relatively narrow interval from 0.005 to 0.01. This was a five- to ten-fold increase in comparison to the stable cell lines.

These observations on chromosomally unstable colorectal cell lines have demonstrated several typical patterns of chromosomal numerical abnormalities, which could be found and were reported for the majority of human solid tumors. Importantly, our modeling results show that all of these patterns could be a result of the same underlying mechanism of random segregation error [29].

4. TRANSITION PROBABILITY MODEL

Using direct computer simulation, we were able to reproduce many observations on the short-term chromosomal changes in cancer cells. Due to the computer memory and speed limitations, the use of computer simulation is currently limited to the modeling of 25–35 cell generations of clonal growth. However, it would be important to be able to analyze long-term trends in chromosome numerical changes in tumors within time intervals from 100 to 1,000 generations. It could provide us with in-depth understanding of the phenomenon of chromosomal instability.

In the long-term observations, many chromosomally unstable tumors and cancer cell lines demonstrate quasi-stable distributions of chromosome copy numbers, which are maintained for hundreds of cell generations. This phenomenon of "stability within instability" presents an interesting paradox which could be address mathematically. However, this requires modeling of long term growth of tumor cell clones for hundreds of generations, which far exceeds capabilities of modern computer systems.

In this study, we have analyzed the long-term behavior of our model using a semianalytical approach. A transition probability matrix was derived analytically and implemented in a recursive algorithm for computational experiments. Using this transition probability model, the expected frequencies of chromosome copy number have been calculated under various initial and boundary conditions.

4.1. Basic Model Assumptions

For this model, we have followed the same basic assumptions of our model of random chromosome segregation errors [25,29]. Each cell in a population contains a nonnegative integer number of discrete objects—chromosomes of several types. Each type is represented in a cell by a number of identical copies. In the model, we describe human cells with 23 types of chromosomes, with normal human cells having two identical chromosomes of each type (diploid female cells), and tumor cells having variable number of chromosomes of each type (aneuploid cells).

In our model, the life cycle of a chromosome includes two types of events: duplication and segregation (Figure 1a). Each chromosome duplicates itself during each cell cycle to produce an identical copy. Segregation occurs at cell division when the two identical copies are separated and transferred into two daughter cells, providing one chromosome copy for each cell. The model describes segregation error as a random event involving the transfer of both copies into the same daughter cell. In the model, we assume that all chromosomes act independently and have equal probability of segregation error, p .

According to our model, each pair of copies is divided randomly between two daughter cells. A segregation event has three possible outcomes (Figure 1a): the first daughter cell receives both copies, the second daughter cell receives both copies, or each of the daughter cells receives one copy (normal segregation). For a given p , we calculate the probability of each outcome as follows:

- (1) with probability $0.5p$, both copies transfer to the first daughter cell;
- (2) with probability $0.5p$, both copies transfer to the second daughter cell;
- (3) with probability $1 - p$, a pair of copies segregates properly (one to each daughter cell).

It is important to note that a single segregation error affects both daughter cells. One of the daughter cells receives two copies of a particular chromosome and the other receives none.

In the model, loss or gain of chromosomes can only occur as a result of improper segregation at division; i.e., chromosomes are not created *de novo* and are not eliminated from the live cells (conservation law for the individual chromosomes). No migration between the types of chromosomes is allowed in the model (conservation law for the chromosome types).

4.2. Computational Algorithm

Let us assume that chromosomes of only one type may have segregation error with probability p . Under this assumption, we can consider only chromosomes of this type. Temporarily, we also assume that cells survive even without chromosomes of this type.

Let us consider a set of vectors S_n consisting of values s_{jn} , average numbers of cells having j chromosomes in generation n ; $j = 0, 1, 2, \dots$; $n = 0, 1, 2, \dots$. Also, we consider matrix P consisting of P_{ik} , expected number of cells with k chromosomes in generation $n + 1$, under the condition that there is just one cell in generation n and it has i chromosomes. Evidently, P_{ik} values are independent on n . These values depend on the probability of segregation error, p , only.

Under these assumptions, we start with one cell having a given number of chromosomes, i.e., initial distribution S_0 . For instance, in our modeling of normal cells, we start with one cell having two chromosomes; i.e., we define initial distribution as

$$S_0 = \{0, 0, 1, 0, \dots\}.$$

Then distributions in sequential generations of cells can be calculated as

$$\begin{aligned} S_1 &= PS_0, \\ S_2 &= P^2 S_0, \\ &\vdots \\ S_n &= P^n S_0. \end{aligned} \tag{1}$$

Notice that

$$\begin{aligned} \sum_{j=0}^{\infty} P_{ij} &= 2, \\ \sum_{j=0}^{\infty} S_{in} &= 2^n \sum_{j=0}^{\infty} S_{i0}. \end{aligned} \quad (2)$$

It is clear from formula (1) that the distributions for any generation, S_n , can be easily determined through the initial distribution, S_0 , as soon as a transition probability matrix P is calculated.

4.3. Transition Probability Matrix

Let us consider a cell having i chromosomes. A probability that exactly k chromosomes will have segregation errors, R_{ik} is equal to $\binom{i}{k} p^k (1-p)^{i-k}$, where $\binom{i}{k}$ is a binomial coefficient, $0 \leq k \leq i$;

$$R_{ik} = \binom{i}{k} p^k (1-p)^{i-k}. \quad (3)$$

Let us define a segregation error as being "positive" if the first daughter cell gets both chromosomes, and being "negative" if the second daughter cell gets both chromosomes. The positive and negative segregation errors may partly compensate each other. Evidently, the probability to have m "positive" among k segregation errors, Q_{km} , is equal to $\binom{k}{m} 0.5^m 0.5^{k-m} = \binom{k}{m} 0.5^k$;

$$Q_{km} = \binom{k}{m} 0.5^k. \quad (4)$$

We assume that $Q_{km} = 0$ for $m < k$. Let us define the "shift", t , i.e., difference between "positive" and "negative" segregation errors. If the initial number of chromosomes is i , number of errors is k , and the number of "positive" errors is m , the shift $t = 2m - k$ (t can be negative). For any m if k is odd or even, then t is odd or even, too. If $t \neq 0$, then there is one cell with $i + t$ chromosomes and one cell with $i - t$ chromosomes. If t is equal to 0, there are two daughter cells with i chromosomes.

The probability T_{it} to have the shift t from the initial number of chromosomes i is equal to

$$T_{it} = \sum_{\text{odd/even } k=0}^i R_{ik} * Q_{k((k-t)/2)}.$$

This sum includes odd or even values of k depending on if t is odd or even, and

$$T_{it} = 0, \quad \text{if } t > i \text{ or } t < -i.$$

Since the values Q_{km} are defined and positive for $0 \leq m \leq k$ only, the terms of this sum are positive when $0 \leq (k+t)/2 \leq k$ or $-k \leq t \leq k$ or $|t| \leq k$. Therefore, we can rewrite this expression as

$$T_{it} = \sum_{\text{odd/even } k=|t|}^i R_{ik} Q_{k((k-t)/2)}. \quad (5)$$

Now, let us calculate P_{ij} ($j \leq 2i$), i.e., average number of cells with k chromosomes, under the condition that there was one cell with i chromosomes. If $i = j$, this means that the shift $t = 0$ and both daughters have the same numbers of chromosomes, so

$$P_{ii} = 2T_{i0}. \quad (6)$$

If $i \neq j$, one daughter cell with j chromosomes will be produced from the mother cell with i chromosomes when $t = j - i$ or $t = -(j - i) = i - j$, and so

$$P_{ij} = T_{i(j-i)} + T_{i(i-j)}. \quad (7)$$

Evidently, (6) is just a special case of (7). Substituting (5) into (7),

$$P_{ij} = \sum_{\text{odd/even: } k=|i-j|}^i R_{ik} Q_{k((k+i-j)/2)} + \sum_{\text{odd/even: } k=|j-i|}^i R_{ik} Q_{k((k+j-i)/2)}$$

or

$$P_{ij} = \sum_{\text{odd/even: } k=|j-i|}^i R_{ik} (Q_{k((k+i-j)/2)} + Q_{k((k+j-i)/2)}). \quad (8)$$

This sum includes odd or even values of k only depending on whether $(i - j)$ is odd or even. Substituting (3) and (4) into (8), we get the following expression:

$$\begin{aligned} P_{ij} &= \sum_{\text{odd/even: } k=|j-i|}^i \binom{i}{k} p^k (1-p)^{i-k} \left(\left(\frac{k+i-j}{2} \right) 0.5^k + \left(\frac{k+j-i}{2} \right) 0.5^k \right) \\ &= \sum_{\text{odd/even: } k=|j-i|}^i \binom{i}{k} p^k (1-p)^{i-k} 0.5^k \left(\left(\frac{k+i-j}{2} \right) + \left(\frac{k+j-i}{2} \right) \right). \end{aligned} \quad (9)$$

Notice that

$$\begin{aligned} \left(\frac{k+j-i}{2} \right) &= \frac{k! / ((k+j-i)/2)!}{(k - ((k+j-i)/2))!} \\ &= \frac{k! / ((k-j-i)/2)!}{(k+i-j)!} = \left(\frac{k}{k+i-j} \right). \end{aligned}$$

Substituting this result into (9), we finally have

$$P_{ij} = 2 \sum_{\text{odd/even: } k=|j-i|}^i \binom{i}{k} p^k (1-p)^{i-k} 0.5^k \left(\frac{k}{k+i-j} \right). \quad (10)$$

Equations (1) and (10) allow a fast recursive calculation of the chromosome distribution for as many cell generations as needed. The only limitation for this algorithm is the size of the matrix P . In order to restrict the size of this matrix, we assumed that the number of chromosomes of a single type could not exceed some large value, $N_{\max}$. In our calculation, we used $N_{\max} = 500$ that is far in excess of the maximum number of chromosome copies known from the experiments. In addition, changes of $N_{\max}$ (e.g., from 500 to 1,000) do not effect the results of calculations up to several hundreds generations.

4.4. Results of Computational Experiments—Basic Model

The long term dynamics of chromosome number distribution has been analyzed using this fast recursive algorithm. We conducted a series of computational experiments, and the expected frequencies of the chromosome copy number have been calculated under various initial and boundary conditions. Although initially we derived a transition matrix corresponding to our basic model

with constant probability of segregation errors, the same approach has been successfully implemented for several alternative models of segregation errors (see Sections 4.5 and 4.6).

In each experiment, we defined the initial distribution S_0 corresponding to one initial cell having a given number of copies of one particular type of chromosome. For instance, in our modeling of clones of normal cells, we start with one cell having two chromosomes, i.e., initial distribution $S_0 = \{0, 0, 1, 0, \dots\}$. The input parameters included initial distribution S_0 , constant probability of segregation errors p , and number of generations n . The program generated average frequencies of chromosome copy number as a function of the number of generations within a single cell derived clone.

The standard statistics were also calculated for every cell generation. Several most informative statistics such as average, standard deviation, mode, and total percentage of cells with nonmodal number of chromosomes (percentage of off-mode cells) were derived. Those statistics were used to compare the outcomes of different models with various qualitative and quantitative experimental observations.

Let us consider the results of simulation for up to 200 generations for several typical values of p which lie within the range corresponding to real experimental observations. Some of the properties of these distributions cannot be derived without an explicit analytical model. But using our computational model, we have been able to obtain some important features of these distributions which could be directly compared with the experimental data (Figures 3–7).

4.4.1. Normal cells: Low rates of segregation errors ($p = 0.001$)

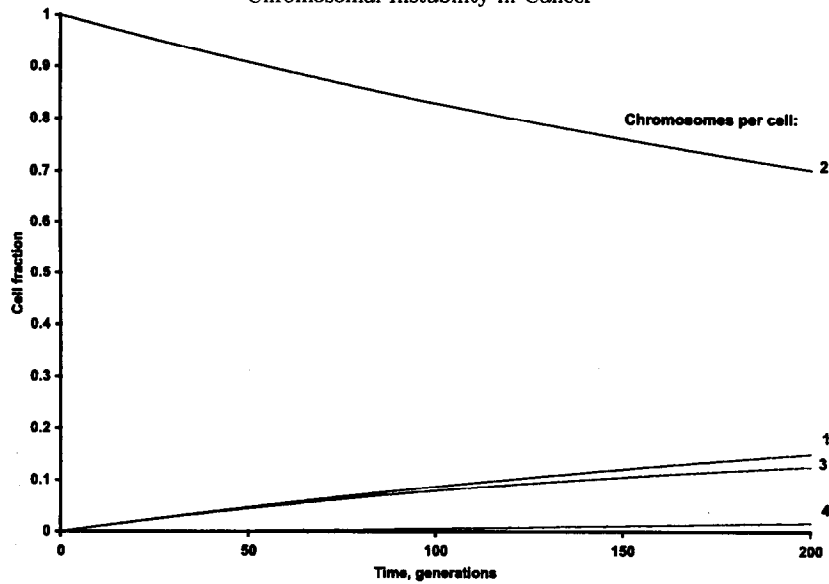
For $p = 0.001$, we continued calculations up to 500 generations that correspond to several years of exponential growth. As a result, we obtained distribution of chromosome copy numbers which is quasi-stable (i.e., stable under condition of elimination of cells with zero copies of chromosome) with a constant mode equal to the initial chromosome number (Figure 3). For comparison with *in vivo* experimental observations, one should keep in mind that in normal tissues, the maximum number of generations for stem cells is within a range of 100 generations over the lifetime of the organism. Also, the modeling of the dynamics of “production” of cells with nonmodal copy number was in a good agreement with *in vitro* experimental observations on stable cell lines (Figure 7, data published in [20] and analyzed in [29]).

4.4.2. Chromosomally unstable cells: Moderate rates of segregation errors ($p = 0.01$)

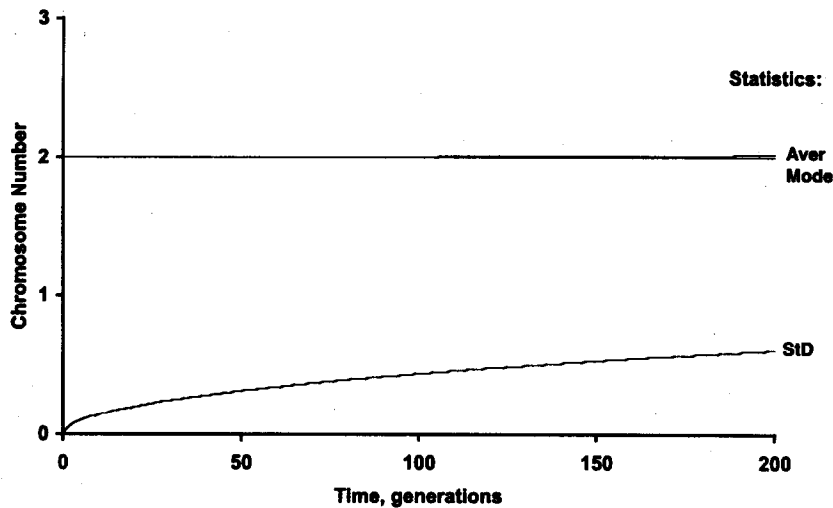
For $p = 0.01$, we continued calculations up to 200 generations that correspond to at least a year of clonal growth. This is a fast growing population with moderate rate of segregation errors (Figure 4). Up to generation 100, the mode is stable and equal to the initial chromosome copy number. After that, it decreases toward a lower number of copies and eventually reaches one (minimal allowable state for an individual cell). The range of values is quite limited with 99% cells having less than eight chromosomes for generation 200. The final distribution is quasi-stable. The dynamics of the initial 50 generations is in a good agreement with the data on the cell lines with CIN phenotype [20], where clones were grown for an average of 25 generations continuously from the single cells, then for up to total of 50 generations (with intermediate dilution and recloning of primary clones) [20]. The modeling of dynamics of “production” of cells with nonmodal copy numbers was in a good agreement with experimental observations on unstable colorectal cancer cell lines (Figure 7, data published by [20] and analyzed in [29]).

4.4.3. Chromosomally unstable cells: High rates of segregation errors ($p = 0.1$)

For $p = 0.1$, we continued calculations up to 200 generations that might correspond to at least a year of clonal growth. For this high rate of segregation errors, the population growth is known to be slow (due to significant fraction of nonviable cells) but stable. The survival fraction is close to one. After a few dozens of generations, the distribution becomes quasi-stable with mode at one



(a) Dynamics of the average frequencies of chromosome numbers.



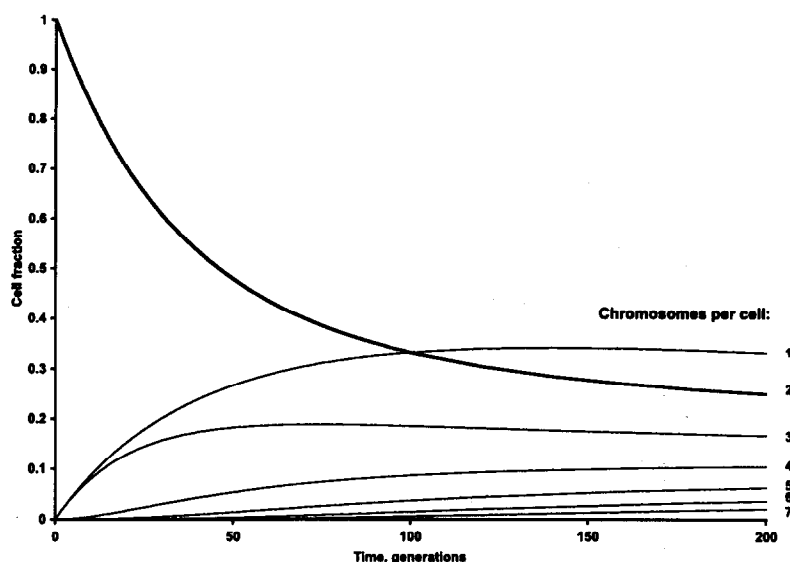
(b) Statistics of chromosome number distribution: average, mode, and standard deviation.

Figure 3. Results of computational experiments. Basic model with $p = 0.001$, $N_{ini} = 2$, corresponding to a normal cell.

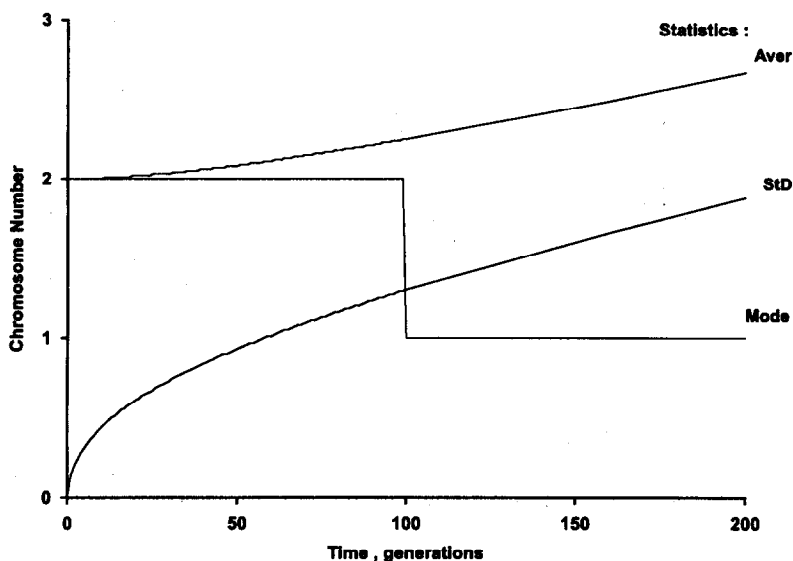
(Figure 5, initial number of chromosomes equals four). These stable frequencies do not depend on initial conditions: clones generated by cell with any initial number of chromosomes reaches the same stable distribution which can be approximated as exponential for the given number of generations (Figure 8). The average number of chromosomes and the range of frequencies become stable within limited number of generations. The percentage of cells with nonmodal copy number is very high and reaches constant saturation value even after 25 generations. Such a rate of chromosomal errors is much higher than the rates which were observed in cell culture and could be similar to heterogeneous cell populations in some solid tumors (Figure 7).

4.4.4. Chromosomally unstable cells: Critical rates of segregation errors ($p = 0.25$)

For $p = 0.25$, the rate of segregation errors reaches a critical value for which only less than one percent of cell population can survive. The dynamics of distribution of copy numbers is



(a) Dynamics of the average frequencies of chromosome numbers.

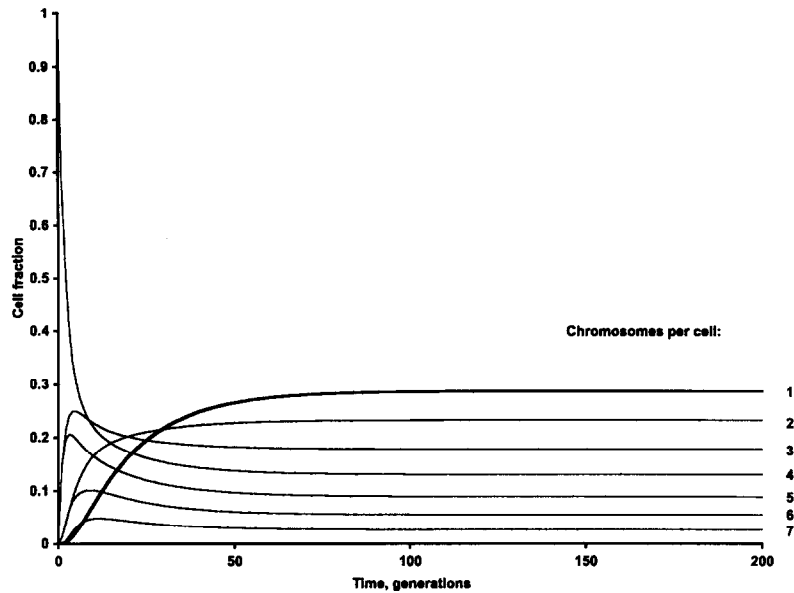


(b) Statistics of chromosome number distribution: average, mode, and standard deviation.

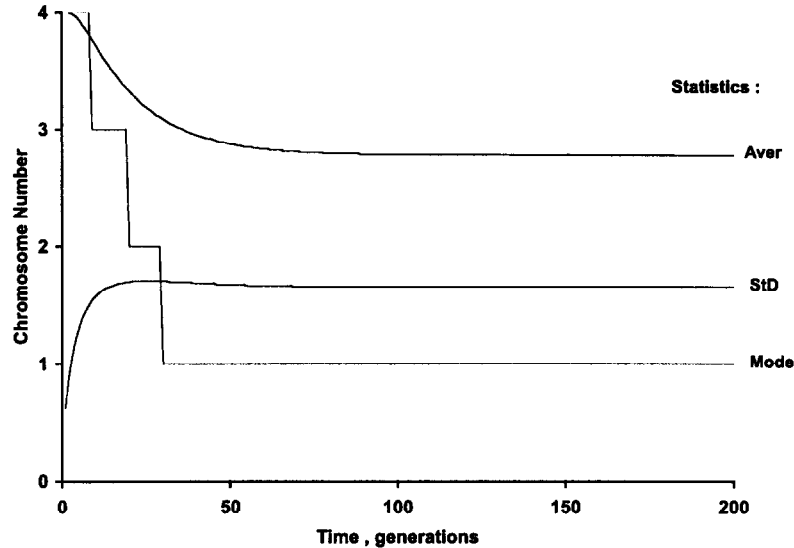
Figure 4. Results of computational experiments. Basic model with $p = 0.01$, $N_{\text{ini}} = 2$.

significantly different than that for the lower p values (Figure 6, initial number of chromosomes equals two).

The modal frequency is changing nonmonotonically with almost immediate (after the third generation) shift of mode to one copy (Figure 6b). The mode then increases back to two copies within the limited number of generations (after 60 generations), and later on (after 300 generations)—to three copies with small difference between the numbers of cells with two and three. For this value of p , the tail of the distribution is very long, and the average number of chromosomes is within the range of ten to 30 copies. Such a rate of chromosomal errors is much higher than the rates which were observed in cell culture and in tumors. Such a rate could be present only for the limited number of generations during critical early stages of malignant transformation and/or during some catastrophic changes in the tumor microenvironment. This might be evident by the



(a) Dynamics of the average frequencies of chromosome numbers.



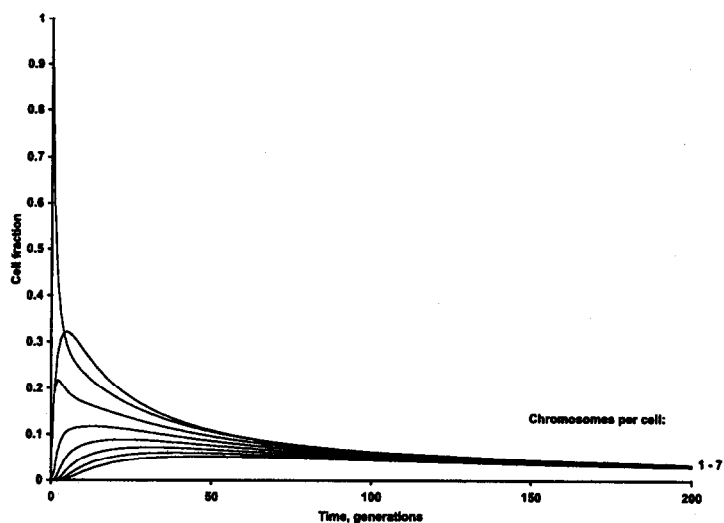
(b) Statistics of chromosome number distribution: average, mode, and standard deviation.

Figure 5. Results of computational experiments. Basic model with $p = 0.1$, $N_{ini} = 4$.

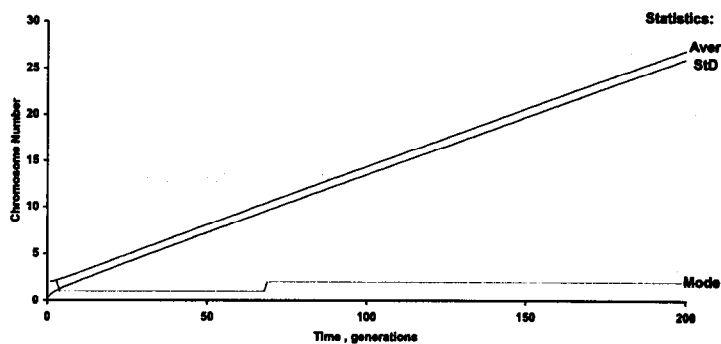
patterns of clonal distributions of chromosome numbers in tumors where modal frequencies are often observed to be within range from one to three copies for different chromosome types within the same tumor [10,12,32]. This heterogeneity of chromosome number distributions could be a result of the earlier episodes of significant increase of rate of segregation errors.

4.4.5. Discrepancies between long term outcomes of the basic model and experimental data

Using our basic model, we assumed constant rate of segregation errors and no limitations on the maximum number of chromosome copies. We found asymptotic quasi-stability of chromosome number distributions being a consequence of the considered p -values. However, there is a



(a) Dynamics of the average frequencies of chromosome numbers.



(b) Statistics of chromosome number distribution: average, mode, and standard deviation.

Figure 6. Results of computational experiments. Basic model with $p = 0.25$, $N_{ini} = 2$.

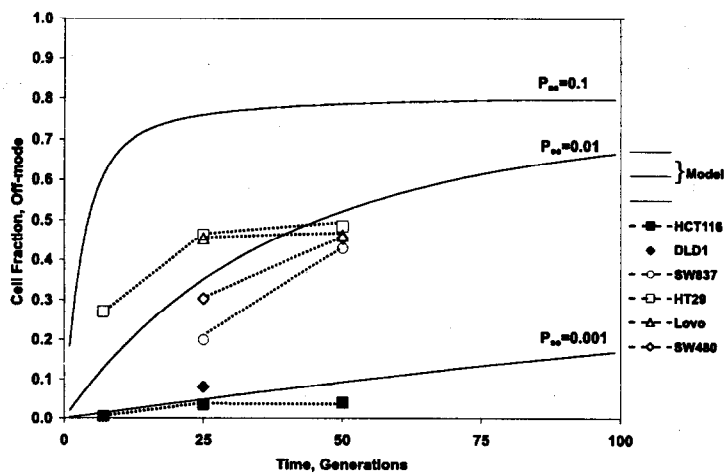


Figure 7. Modeling results (solid lines) versus experimental data (symbols). Single cell derived clones of stable (filled symbols) and unstable (open symbols) colorectal cell lines were analyzed by *in situ* hybridization with 12 chromosome specific probes. Average fraction of cells with nonmodal chromosome numbers were calculated after eight, 25, and 50 generations based on published data [20].

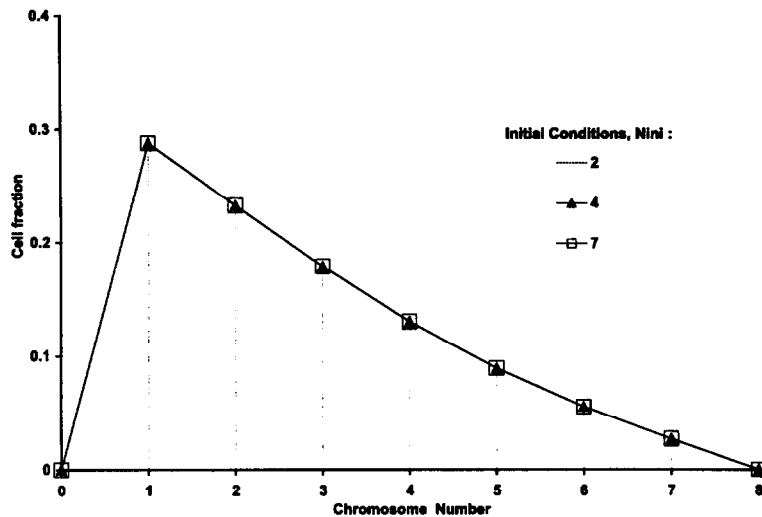
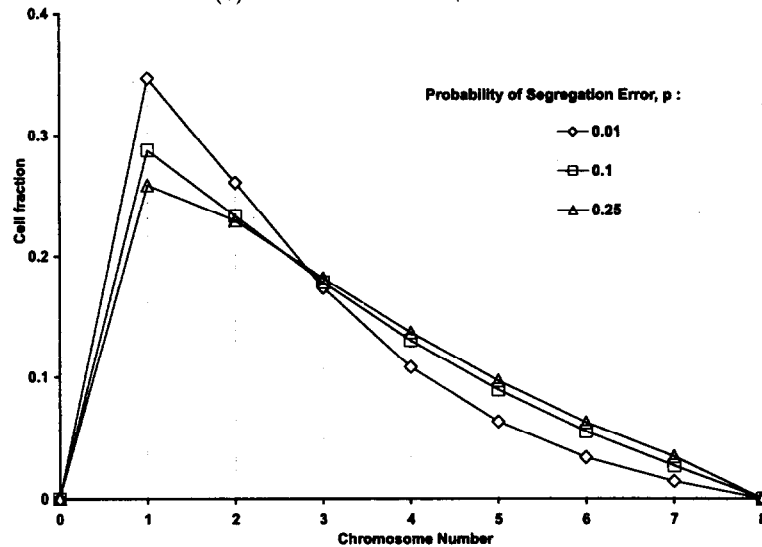
(a) Constant rate model, $p = 0.1$.(a) Stable distribution of chromosome number (500 generations) for $p = 0.1$. Three sets of data are shown for $N_{ini} = 2, 4, 7$.(b) Constant rate model, $N_{ini} = 2$.(b) Stable distribution of chromosome number (500 generations) for $N_{ini} = 2$. Data are shown for $p = 0.01, 0.1, 0.25$.

Figure 8. Basic model of segregation errors (constant probability of segregation errors model). Stable distribution of chromosome numbers does not depend on initial conditions (number of chromosomes in the initiator cell, N_{ini}) and has weak dependence on probability of segregation errors, p .

disagreement of some of our computational results with some of the experimental data on the distributions of chromosome copies number.

- a. Modal value for the distributions of a copy number for a single type of chromosome. In experiments, especially with cell cultures, the mode of the chromosome number distributions lies within a range from one or six (with two, three, and four copies being more common, although one copy is also seen often). The basic model demonstrated an asymptotic distribution with maximum frequency at smallest allowable state, i.e., with mode at one chromosome copy per cell.

- b. Tails of the distribution. In many experimental observations, the maximum number of chromosomes of the same type does not exceed eight to 12 copies. The basic model demonstrated longer tails of chromosome copy distribution.

Both discrepancies can be explained within the basic model by the assumption of a relatively small value of p . This assumption does not appear to be unrealistic. Both properties of the basic model: shift of the mode to one copy and the long tails cannot be observed if the rates of segregation errors does not exceed 1% per chromosome. This is consistent with our previous results [25] which indicate that the plateau of 100% survivable clones exists only for the rates of segregation errors which are less than 1% percent per chromosome.

This assumption, that for many cell lines rate of segregation errors is small ($p < 0.01$), explains the modes and provides a reasonable estimation of dispersions (off-mode cell counts) which were obtained for several colorectal cancer cell lines (Figure 7).

4.5. Additional Assumptions for a Modified Model of Chromosome Segregation Errors

In our basic model, we have assumed a constant probability of segregation errors which has clear biological meaning of constant average rate of segregation errors. Such a basic model reproduces short-term observations and can provide a reasonable explanation of some long-term observations. However, it would be useful to test some additional assumptions to explain some specific differences between this simple constant probability model and data. Such differences are not surprising for the basic model having, in fact, only one adjustable parameter—probability of segregation error, p .

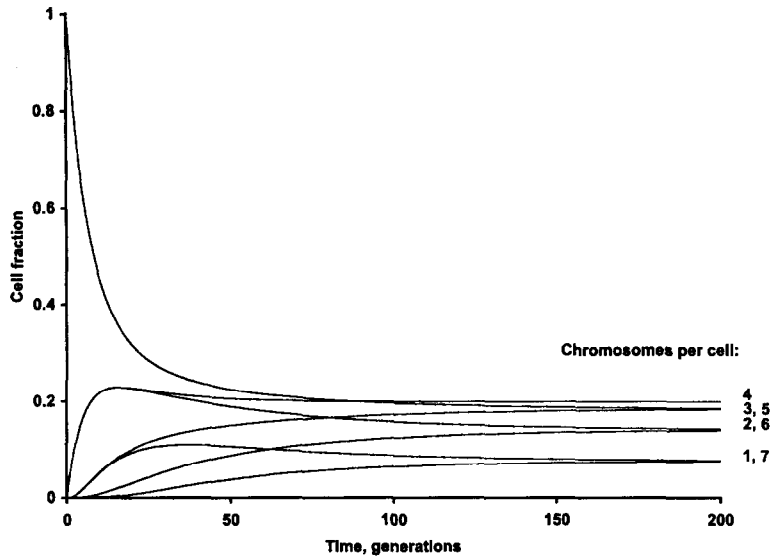
4.5.1. Limitations on the maximum number of chromosomes

As a first step, we have included a simple assumption of a limit on the maximum number of chromosomes, $N_{\max}$, which a cell could tolerate. In this modified model, we describe an upper absorption boundary for the number of chromosome copies for each type of chromosome. For any type of chromosome, we introduce conditional cell loss if the number of copies $N \geq N_{\max}$. This assumption provides a mechanism which stabilizes the tails of the distribution. The presence of such a limitation became evident when we compared our simple one-parameter model to the data on clonal distributions of high ploidy cell lines [20,29]. Using this new assumption, we have adjusted our recursive algorithm and have tested this modified model in computational experiments. We have tested various values of the maximum number of chromosome copies, $N_{\max}$, within the range of six to 12 copies. Naturally, results of computational experiments have shown that these distributions do not have expanding tails. All of them tend to be stable after some transient process within a limited number of generations (see Figure 5). Nevertheless, all of them, in a long run, have produced stable distribution with the mode at one copy, and the near exponential shape of distribution. For this model, the final distribution of copy numbers does not depend on initial conditions (Figure 8a), and has a weak dependency on the rate of segregation errors (Figure 8b), and values of $N_{\max}$.

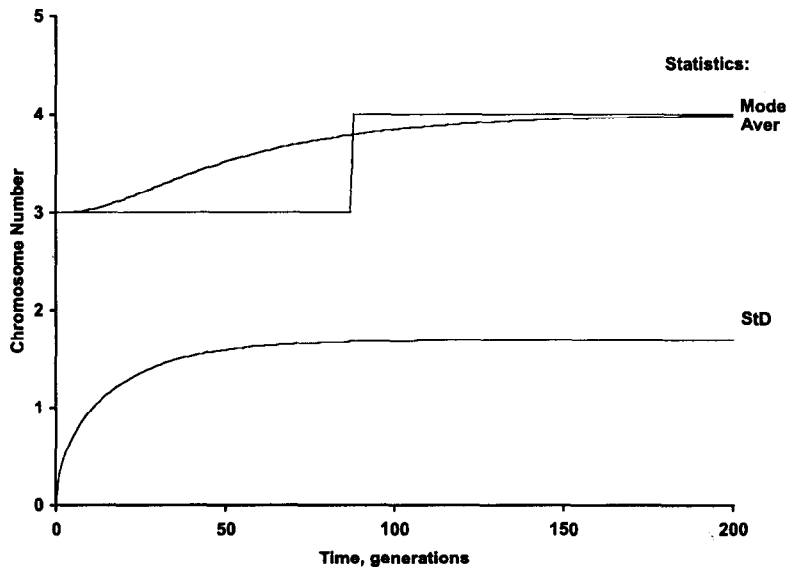
Nevertheless, it is interesting to investigate other possibilities to explain the experimental data distribution. In this case, the most serious disagreement is the quasi-exponential shape of distribution with the modal frequency at one copy.

4.5.2. Limitations on the number of random segregation errors

Our basic model has included an assumption of constant probability of segregation errors which corresponds to a constant average rate of segregation errors per cell division. However, this assumption provides a very specific pattern of actual numbers of events per chromosome type.



(a) Dynamics of the average frequencies of chromosome numbers.

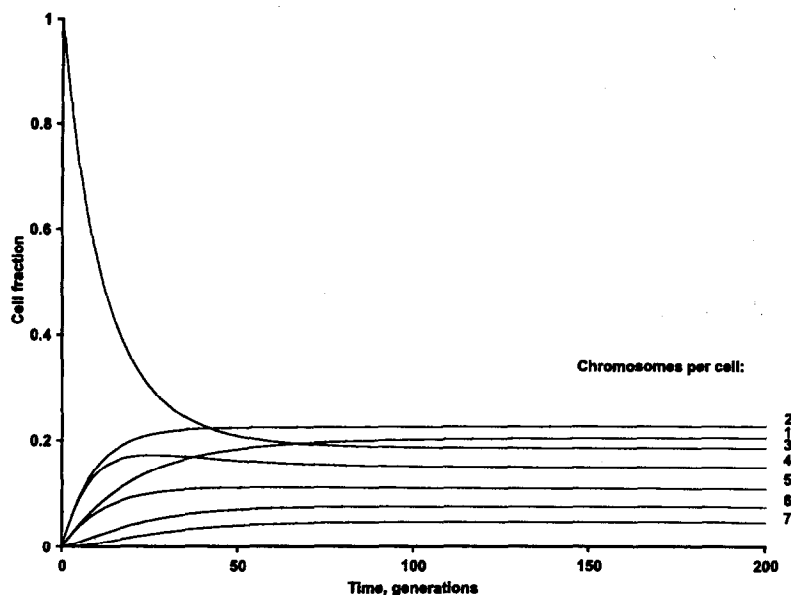


(b) Statistics of chromosome number distribution: average, mode, and standard deviation.

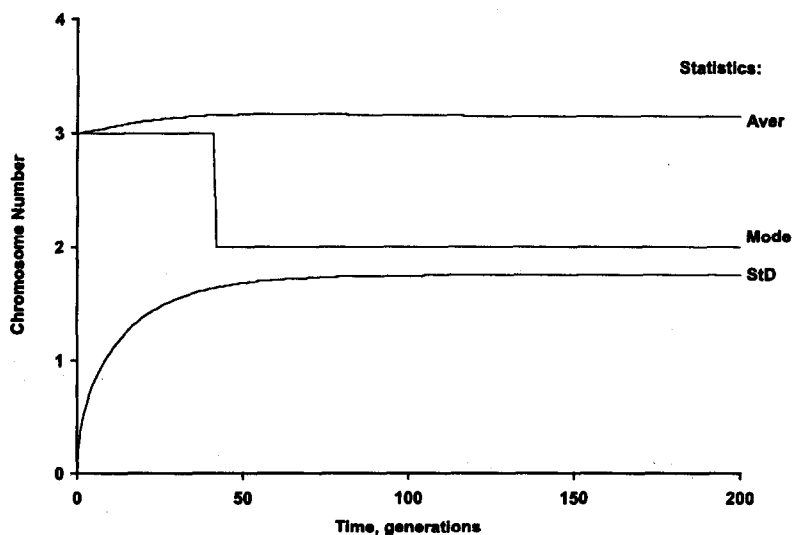
Figure 9. Alternative model of segregation error mechanism: single error per cell, ($p = 0.1$, $N_{ini} = 3$).

For each type of chromosome, the average number of segregation errors in a cell is proportional to the actual number of the chromosomes of this type within a cell. Therefore, the actual number of segregation errors will be higher for those cells which have more copies of a chromosome. This results in the specific form of the asymptotic distribution of chromosome copies with the maximum frequency (mode) at the minimal permissible copy number. Such asymptotic behavior has been shown analytically for the model of unstable gene amplification [27].

To provide an alternative to this specific assumption, we have investigated a different modeling assumption on the number of segregation errors. We assumed that only one segregation error among chromosomes of particular type is possible during a single division. Under this assumption, the probability of segregation error p defines a probability of single event of segregation error per cell division.



(a) Dynamics of the average frequencies of chromosome numbers.



(b) Statistics of chromosome number distribution: average, mode, and standard deviation.

Figure 10. Alternative model of segregation error mechanism: abnormal mitosis ($p = 0.1$, $N_{ini} = 3$).

For this model, instead of formula (10) we have

$$P_{ij} = 0, \quad |j - i| > 1, \quad (11.1)$$

$$P_{ij} = 2(1 - p), \quad j = i, \quad (11.2)$$

$$P_{ij} = p, \quad |j - i| = 1. \quad (11.3)$$

As in the previous set of calculations, we assumed that the number of chromosomes of the same type is limited by N_{max} .

In this case, the distributions also tend to be stable, but the shape and the mode of distribution is different (Figures 9 and 11). Moreover, the stable distribution strongly depends on N_{max} , and

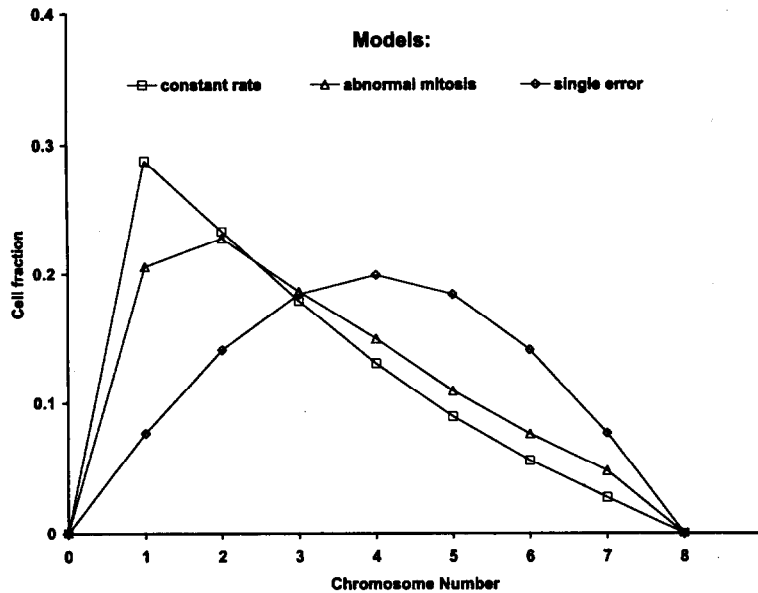


Figure 11. Stable distributions of chromosome number. Three alternative models of mechanisms of segregation error: standard model (constant probability); single error and abnormal mitosis. Note the differences of the modal chromosome numbers.

the dependence of the final distribution on the probability of nonseparation is very weak (this can be proven mathematically).

For instance, for $N_{\max} = 8$, the asymptotic distributions are symmetrical with mode at four (Figure 9). This result is quite interesting because a tetraploid mode (four copies of each chromosome type) is commonly observed in many tumors and cancer cell lines.

For such a model, a change in the upper limit produces a change in the position of the mode. For instance, for the maximum number of copies $N_{\max} = 4$, this model generates the symmetric distribution with mode at two.

Up to this point, we have tested most of the evident modifications of the basic model. Yet these modifications were unable to reproduce some experimental data. This suggests that the rate of segregation errors is not so low as our basic model required.

4.6. Alternative Model of Chromosome Segregation Errors: Abnormal Mitosis

We have constructed an alternative model of segregation errors. For this model, we assumed that there are two types of cell division.

1. Normal division—when all chromosome pairs segregate properly and each of daughter cells receives exactly one chromosome of any pair.
2. Abnormal division—when each chromosome of a pair is randomly distributed between the daughter cells.

Once again, we introduce single parameter $-p$, probability of abnormal mitosis, which is assumed to be constant, and represents an average rate of abnormal mitosis. We have derived a transition probability matrix for this model and have considered two versions of this model: with restriction on the maximum number of copies and without this restriction.

For this model, instead of formula (10) we have

$$P_{ij} = 0, \quad j > 2i, \quad (12.1)$$

$$P_{ij} = 2(1-p) + 2p(0.5)^{2i} \binom{2i}{j}, \quad j = i, \quad (12.2)$$

$$P_{ij} = 2p(0.5)^{2i} \binom{2i}{j}, \quad j \leq 2i, \quad j \neq i. \quad (12.3)$$

Computational experiments demonstrated that this model has an asymptotic distribution of chromosomes with a stable mode of two copies (Figure 10), which does not depend on initial number of chromosomes, the probability of abnormal mitosis, or the maximum number of chromosome copies.

This new model provides valuable addition to the set of models presented in this study, as it shows stable diploid mode (two copies) in a presence of large scale abnormal mitotic events.

5. CONCLUSION

In long-term observations, many chromosomally unstable tumors and cancer cell lines demonstrate stable distribution of chromosome copy numbers, which are maintained for hundreds of cell generations. This phenomenon of “stability within instability” presents an interesting paradox which could be addressed mathematically. In this study, we have developed a set of models of chromosomal instability which were designed for quantitative analysis of long-term dynamics of chromosome numerical changes in cancer. We have investigated asymptotic behavior of these models using a semianalytical approach. A transition probability matrix was derived analytically for each type of model and implemented in a recursive algorithm for computational experiments. Using these transition probability models, the expected frequencies of chromosome copy number have been calculated under various initial and boundary conditions (Figures 3–6, 8–10). We have tested several alternative models which describe various mechanisms of chromosome segregation errors, and found conditions of stabilization of distribution of chromosomes copy numbers over large number of cell generations. Stable clonal frequencies were estimated which are independent of initial conditions (i.e., chromosome copy numbers in the initiator cells) as well as of probability of segregation errors (Figure 8). These stable distributions were, however, dependent on the model assumptions regarding particular mechanism of mal-segregation of chromosomes (Figure 11). Thus, our modeling results suggest a possible connection between the form of stable distributions of chromosome numbers in tumors and the underlying mechanism of chromosome mal-segregation. This new analytical approach allows to overcome technical impairments and limitations of computer simulation, and provides mathematical insight into possible scenarios of long-term evolution of chromosome numerical changes in human tumors.

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